



EDUCATION

University of California San Diego <i>Ph.D., Electrical & Computer Engineering</i> San Diego, CA	July 2025 – Present
University of California San Diego <i>M.S., Electrical & Computer Engineering</i> San Diego, CA	September 2023 – June 2025
Birla Institute of Technology and Science <i>B.E., Electronics & Communication Engineering</i> Hyderabad	August 2016 – June 2020

RESEARCH EXPERIENCE

Wireless and Battery-Free Universal Sensing Platform (<i>Patent Pending</i>)	April 2024 – Present
• PhD student (advisor: Prof. Dinesh Bharadia) building wireless, battery-free sensing systems; 5 papers, a Best Paper and 2 Best Demo awards. • Built SenSync, a time-alignment and phase-correction algorithm for RFID sensing: 5x sensory resolution, 0.79° median phase error. • Designed TuneTag, an impedance-matched sensing tag: 5x sensing accuracy, 2.4x range, and 5x faster sub-second response vs. state of the art. • Co-developed ZenseTag, a 15x10 mm flexible passive tag reading force, soil-moisture, and light sensors (93%+ moisture accuracy). • Developed an Augmented Reality smartphone app that auto-detects sensors and overlays real-time readings on a live camera feed. • Raised RFID read throughput 8x with a custom data-collection platform on the Impinj Octane SDK, enabling real-time sensing. • Earned all 5 ACM artifact evaluation badges (available, functional, reusable, reproduced, replicated) for ZenseTag at SenSys '24.	

PUBLICATIONS

- **I. Bansal** et al., “*TuneTag: Impedance-Matched RFID Tag for Long-Range and Reliable Battery-Free Sensing*,” IEEE Journal of RFID, Vol. 9, 2025.
- **I. Bansal** et al., “*SenSync: Real-Time and Accurate Passive Sensing*,” IEEE RFID 2025. **Best Paper Award**.
- N.B., A.G., **I. Bansal** et al., “*ZenseTag: An RFID assisted Twin-Tag Single Antenna COTS Sensor Interface*,” ACM SenSys 2024.
- **I. Bansal** et al., “*Demo - SIGAR: Sensor Integration Gateway using Augmented Reality*,” ACM SenSys 2025. **Best Demo Runner-Up**.
- **I. Bansal** et al., “*Demo - ZenseTag: Real-Time Passive RFID Sensing*,” ACM MobiCom 2024. **Best Demo Runner-Up**.

HONORS & AWARDS

- **Winner**, ARA Hackathon (Arathon) at the NSF AgRuralG Workshop (AgWireless '26) for rural cellular infrastructure placement – 2026.
- **Best Paper Award**, IEEE RFID 2025 for SenSync, the algorithm behind the battery-free sensing platform – 2025.
- **Best Demo Runner-Up**, ACM SenSys 2025 (SIGAR) and ACM MobiCom 2024 (ZenseTag) – 2024, 2025.
- **Centaur Graduate Student Fellowship**, UC San Diego, for outstanding contributions to research and development – 2025.
- **Employee of the Quarter**, Cisco Systems, for exceptional contributions to new feature development – Q1 FY2021.

PROFESSIONAL EXPERIENCE

Senior Software Developer <i>Cisco Systems</i> Bangalore, India	January 2020 – July 2023
• Built install and upgrade software for Cisco HyperFlex in Scala, Go, and Python, managing public and private cloud infrastructure. • Owned the security and user-experience features of the HyperFlex upgrade software end to end. • Mentored 3+ engineers, including junior developers and newly hired senior engineers. • Cut HyperFlex upgrade time by 22% by optimizing the upgrade software architecture. • Resolved 93% of assigned defects, cutting upgrade-software bugs by 70% and improving cluster lifecycle management. • Improved deployment efficiency by 25% by adopting infrastructure as code with Terraform and Ansible. • Embedded automated testing and continuous monitoring in DevOps pipelines, enabling rapid and reliable software releases. • Engineered automation tools to manage test scripts and test statistics for Intersight, a cloud-operated infrastructure management platform. • Orchestrated containerized core infrastructure for Intersight on Kubernetes, improving scalability, efficiency, and resilience. • Saved the QA team 100 work hours per week by automating test-script review with a tool integrated into the CI/CD pipeline. • Added real-time monitoring and alerting to the CI/CD pipeline with Elasticsearch, Grafana, and Kibana, enabling proactive issue resolution. • Designed a web application that tracks verification activities for hundreds of Kubernetes-orchestrated containerized microservices. • Automated detection of security vulnerabilities and user-experience flaws, saving the QA team 50 work hours per week. • Launched a chatbot for real-time reporting of ongoing and past verification activities, cutting team-wide data retrieval by 40 work hours per week.	

INTERNSHIPS

Summer Intern <i>Peco Pallet Inc</i> New York, NY	June 2024 – September 2024
• Built a Python and Excel pipeline for data cleaning, geocoding, and reporting on large corporate datasets. • Engineered a one-click tool that analyzes large datasets and generates custom reports to estimate optimal pricing strategies. • Improved data quality and cut data-management effort by 70% by building the application in-house. • Sped up pricing delivery by 40%, increasing the probability of conversion by 25%.	
Summer Intern <i>Western Digital</i> Bangalore, India	May 2019 – July 2019
• Built a functional code-coverage tool for firmware verification of removable flash-based storage devices. • Designed a compact data structure and algorithm to compute and store coverage results within a 50 kB on-disk budget. • Added a user-friendly interface that displays results and suggests actions to improve test coverage. • Saved an estimated US\$50,000 annually by building the tool in-house.	
Research Intern <i>Indian Meteorological Department</i> Pune, India	May 2018 – July 2018
• Developed an IoT system that streams real-time meteorological data from an Automatic Weather Station to a mobile app. • Designed a primary-secondary topology using Raspberry Pi and Arduino Nano boards for efficient data collection and processing. • Programmed embedded C sensor interfaces and Python services for real-time data streaming to a cloud server. • Integrated temperature, humidity, wind, rainfall, and soil-moisture sensors over I2C, UART, and RS-485. • Achieved 100% uptime in harsh environments with LTE, GPS, Wi-Fi, and Ethernet connectivity options.	

PROJECTS

Data-Driven Rural Infrastructure Placement (ARA Hackathon Winner)

August 2026

- **Led a team** to first place in the ARA Hackathon (Arathon Challenge 3) at AgWireless '26, Ames, Iowa.
- Predicted cellular coverage across a **178 km²** rural area from **7,144** drive-test measurements covering only 7% of it.
- Identified a single macro-site placement raising coverage from **44% to 69%** of route-km and from **37% to 59%** of area.
- Outperformed the challenge's baseline placement in **8 of 8** configurations, by a median **1.52x**.
- Ran a **198**-configuration sensitivity sweep across propagation models, asset classes, and service criteria.

Recommender System for an eCommerce based Rental Clothing Store

November 2024 – December 2024

- Built a custom **latent factor model** achieving an MSE of **0.317**, outperforming baseline, **TF-IDF**, and **SVD** models.
- Treated all sizes of an item as one product, reducing data sparsity and improving prediction accuracy.
- Designed and evaluated linguistic-feature and physical-characteristic models of user satisfaction in clothing rentals.
- Delivered a **recommender system** that generates personalized item recommendations and predicted ratings for each user.

Design and Development of Branch Predictors

May 2024 – June 2024

- Implemented a **tournament branch predictor** combining global and local prediction, reaching a **1.48%** misprediction rate.
- Designed a custom predictor combining **gshare** and local prediction, reaching a **1.64%** misprediction rate.
- Optimized hardware budget allocation across predictors, balancing accuracy and storage within **72–128 Kbit**.
- Benchmarked on GCC, ASTAR, H264ref, and NAMD traces, consistently beating a baseline gshare predictor.
- Applied pattern history tables, branch history tables, and meta-predictors to improve **CPU** performance.

Parallelization of Genetic Pairwise Alignment for ClustalW

January 2024 – March 2024

- Applied **wavefront parallelism** to ClustalW's pairwise alignment step, achieving a **1000x** speedup over the serial implementation.
- Developed a **CUDA** algorithm that parallelizes sequence alignments across GPU kernel blocks.
- Integrated the X-Drop heuristic to terminate suboptimal alignments early, further improving efficiency.
- Stored anti-diagonals in **shared memory** and used **parallel reduction** to maximize GPU throughput.
- Reached a **770x** GPU speedup over a multi-threaded CPU implementation by tuning grid and block sizes.

Dual-Band MIMO Circular Patch Antenna Design and Isolation Analysis

January 2019 – May 2019

- Designed a dual-band (3.5 GHz and 4.5 GHz) circular microstrip patch antenna on an FR4 epoxy substrate.
- Achieved S11 of **-38.46 dB** at 3.5 GHz and **-40.72 dB** at 4.5 GHz.
- Implemented and compared four defected ground structure (DGS) isolation techniques for a **MIMO** configuration.
- Attained up to **28 dB** isolation between antenna elements using dual-strip DGS.
- Analyzed S-parameters and radiation patterns in **Ansys HFSS**.
- Optimized the design for compact MIMO wireless communication systems.

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA, Go, Java, Scala, Kotlin, SQL, Bash, JavaScript, HTML/CSS, YAML
ML & GPU: PyTorch, TensorFlow, CUDA, Machine Learning, Deep Learning, Recommender Systems, AR/VR
Backend & Web: Django, Flask, FastAPI, Spring Boot, Hibernate, GraphQL, REST APIs, Express, React, Angular, Flutter, MongoDB
Infrastructure & DevOps: Docker, Kubernetes, Terraform, Ansible, Jenkins, Git, CI/CD, Elasticsearch, Logstash, Kibana, Grafana, Splunk, Linux
Cloud: Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, VMware vSphere, Cisco HyperFlex, Cisco Intersight
Hardware & RF: Raspberry Pi, Arduino, I2C, UART, RS-485, RFID, Ansys HFSS

COURSEWORK

Algorithms, Cloud Computing, Computer Architecture, Computer Networks, Database Management, Data Structures, Digital Signal Processing, Machine Learning, Object Oriented Programming, Operating Systems, Parallel Computing, Recommender Systems, Software Engineering, VLSI System Design

CERTIFICATIONS

Deep Learning Specialization (Coursera); TensorFlow in Practice Specialization (Coursera); AWS Fundamentals Specialization (Coursera); Machine Learning A-Z: AI, Python & R (Udemy)

EXTRA-CURRICULAR ACTIVITIES

- Organized and hosted 50+ guest lectures for students and faculty at BITS Hyderabad from August 2016 to May 2019.
- Conducted financial literacy programs as a Finance Club member at BITS Hyderabad from August 2016 to May 2018.
- Served as Teaching Assistant for the Department of Economics and Finance at BITS Hyderabad from January 2019 to May 2019.
- Avid traveler, having visited over 20 cities across the United States, China, and India in recent years.